MILITARY SPECIFICATION

PROPELLANT, M6
FOR USE IN CARTRIDGE, CANNISTER
76MM, M363

1. SCOPE

1.1 This specification covers one type of propellant designated as M6, for use in the Cartridge, Cannister, 76MM, M363 for the 76MM guns M32 and M48.

2. APPLICABLE DOCUMENTS

2.1 The following documents of the issue in effect on date of invitation for bids or request for proposal form a part of this specification to the extent specified herein.

SPECIFICATIONS

MILITARY

MIL-A-2550	Ammunition and Special Weapons; General Specification for
MIL-I-45208	Inspection Requirements; General Specification for

STANDARDS

MILITARY

MIL-STD-105	Sampling Procedures and Tables for Inspection by Attributes (ABC-STD-105)
MIL-STD-109	Quality Assurance Terms and Definitions
MIL-STD-414	Sampling, Procedures and Tables for Inspection by Attributes
MIL-STD-652	Propellant, Solid for Cannon Requirements and Packing
MIL-STD-1235	Single and Multilevel Continuous Sampling Procedures and Tables for Inspection by Attributes

FSC: 1375

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DRAWINGS

ORDNANCE CORPS

9204458	Cartridge Cannister M363 for 76MM Guns M32 and M48

(Copies of specifications, standards, drawings and publications required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

2.2 Other publications. - The following document forms a part of this specification to the extent specified herein. Unless otherwise indicated, the issue in effect on date of invitation for bids shall apply.

CODE OF FEDERAL REGULATIONS

49 CFR 71-90	Interstate Commerce Commission Rules and Regulations for the Transportation of Explosives and Other Dangerous Articles

The Interstate Commerce Commission Regulations are now a part of the Code of Federal Regulations (1949 Edition and Revisions) available from the Superintendent of Documents, Government Printing Officer, Washington 25, D. C. Orders for the above publications should cite "49 CFR 71-90 (latest revision)."

3. REQUIREMENTS

3.1 Material. - The propellant shall be M6, Type I and conforming to the requirement of Standard MIL-STD-652. Test data shall be reported in accordance with 6.6.

3.2 Ballistics. - The propellant, when fired in the weapon and with the weight of projectile for which it was manufactured, shall comply with the following requirement when tested as specified in 4.3.1.

3.2.1 Weights. - At 70° F, the weight of propellant assessed as required to provide a pressure of 52,000 pounds per square inch (psi) Piezo equivalent shall exceed 78.0 ounces without exceeding the volumetric capacity of the case.

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3.2.2 Pressure. The average maximum pressure shall not exceed 53,200 psi.

3.2.3 Uniformity of pressure. - The standard deviation of the maximum pressures for the lot at 70° F. shall not exceed 675 psi.

3.3 Workmanship. - The best commercial practices shall be used in the formulation of the propellant furnished under this specification and all other applicable documents. The propellant and its standard ingredients shall be protected from the action of direct sunlight and acid funes. No reground propellant or nitrocellulose shall be used in the manufacture of propellant procured under this specification.

4. QUALITY ASSURANCE PROVISIONS

4.1 General quality assurance provisions. - Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified, the supplier may utilize his own facilities or any commercial laboratory acceptable to the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements. Reference shall be made to Standard MIL-STD-109 in order to define terms used herein. The provisions of Specification MIL-A-2550 shall apply.

4.1.1 Contractor inspection system. - The contractor shall provide and maintain an effective inspection system in compliance with the requirements of Specification MIL-I-45208.

4.1.2 Submission of product. - At the time the completed lot of product is submitted to the Government for acceptance the contractor shall supply the following information accompanied by a certificate which attests that the information provided is correct and applicable to the product being submitted:

a. A statement that the lot complies with all quality assurance provisions of the approved current written description of the system.

b. Number of units of product inspected.

c. Results obtained for all inspection performed.

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d. Drawing, specification number and date, together with an identification and date of changes.

e. Certificates of compliance on all material purchased by the contractor when such material is controlled by Government specifications referenced in any of the contractual documents.

f. Number of items in the lot.

g. Date submitted.

The certificate shall be signed by a responsible agent of the certifying organization. The initial certificate submitted shall be substantiated by evidence of the agent's authority to bind his principal. Substantiation of the agent's authority will not be required with subsequent certificates unless, during the course of the contract, this authority is vested in another agent of the certifying organization.

4.1.3 Government verification. - Using the contractor's written quality assurance procedure, this detail specification, the applicable drawings and other contractual documents as a guide, the Government inspector shall verify at unscheduled intervals all quality assurance operations performed by the contractor. Verification will be in accordance with Specification MIL-I-45208 and will be performed to the extent necessary to assure compliance with the contractual requirements. Severity of Government inspection of individual characteristics will be directly related to the seriousness of the classification assigned. In no instance will a characteristic classified "critical" be accepted solely on the basis of the contractor's records.

4.2 Inspection provisions.

4.2.1 Lot formation. - A lot shall consist of one or more batches of M6, propellant produced by one manufacturer, in accordance with the same specification or same specification revision, under one continuous set of operating conditions. Each lot shall consist of that quantity of M6 propellant which has been subjected to the same unit chemical or physical mixing process intended to make the final product homogeneous. Product shall be submitted for inspection in accordance with Standard MIL-STD-105 (or Standard MIL-STD 1235 when applicable).

4.2.2 Testing

4.2.2.1 Ballistics. - *(see 3.2) A sample of 200 pounds of the propellant* shall be randomly selected from the lot for the ballistic test which will be performed at a Government proving ground.

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4.2.2.1.1 Pressure (Major defect). - The lot shall be rejected if the average maximum pressure at the charge recommended exceeds the average maximum limitation specified in 3.2.2.

4.2.2.1.2 *Uniformity of pressure (see 3.2.3)* (Major defect). - If the product of the factor 0.73 times the pressure standard deviation of the test rounds exceeds the applicable requirement, the lot shall be rejected. For the initial production lots a 20-round sample will be used and the factor will be 0.79 (see 6.2, 6.3 and 6.4).

4.2.2.1.3 Retest. - If for any reason the proof director considers that the test conditions have detrimentally affected the test results, he shall declare the results invalid and perform a new test on additional charges as required.

4.2.2.2 Material (Major defect). - The M6 propellant shall be rejected if the material requirements outlined in Standard MIL-STD-652 are not satisfied.

4.3 Test methods and procedures *(see 6.3)*

4.3.1 Ballistic tests. - These tests shall be conducted at a Government proving ground in accordance with the applicable Ordnance Proof Manual and Acceptance Test Procedure. For the evaluation of the initial production lots, a twenty (20) round uniformity series shall then be fired and the mean pressure (Piezo equivalent) and variance shall be calculated. A ten (10) round uniformity series shall be used for all subsequent lots.

4.3.2 Material. - The M6 propellant shall be inspected in accordance with the procedures cited in Standard MIL-STD-652.

5. PREPARATION FOR DELIVERY

5.1 Packing and marking. - The propellant shall be packed and marked in accordance with Standard MIL-STD-652 or drawings 7549033 and 8858577.

6. NOTES

6.1 Ordering data. - Procurement documents should specify the following:

a. Title, number and date of this specification.

b. Process. - Details of the manufacturing process and the equipment used by the contractor will be submitted to the procuring activity in writing prior to commencement of manufacture. Any deviation from this manufacturing process must be submitted in writing to the procuring activity prior to being put into effect.

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6.2 Standard deviation. - The standard deviation will be calculated with (N-1) as the divisor in a standard statistical technique to that shown on Standard MIL-STD-414, Section B, Example B1.

6.3 Factor 0.79 and 0.73. - The factors 0.79 and 0.73 make allowances for the probability that a standard deviation from a twenty (20) and ten (10) round sample respectively exceeds the true standard deviation by chance alone.

6.4 The initial production lots represent the first three consecutive lots which have been produced and accepted.

6.5 Reliability testing

6.5.1 For informational purposes only a ten-round series shall be fired with rounds which were conditioned for a minimum of 24 hours at 125 degrees F. before firing. These rounds shall contain the full charge weight of propellant to which the lot was assessed. Data on the maximum pressures observed should be noted and submitted as directed in paragraph 6.6.

6.5.2 The above informational test shall be conducted only on the initial production lots (see 6.4).

6.6 Submission of test data. - In addition to the normal distribution of records when the propellant is procured by the Department of the Army, one copy of all test data will be forwarded to the:

Commanding Officer
Picatinny Arsenal
ATTN: SMUPA-ND2
Dover, New Jersey 07801

Custodian:
Army-MU

Preparing activity:
Army-MU

Project No. 1375-A826

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